SUPPORTING INFORMATION

Optimizing Vanadium-Catalyzed Epoxidation Reactions: Machine Learning-Driven Yield Predictions and Data Augmentation

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Annex S1 – Model Database

Complete database, featuring individual bibliographical source, and the list of calculated and experimental descriptors is available at the following repository: https://gitfront.io/r/ESA-Vanadium/biMx4obXpmkj/ESA-Vanadium-Yield/. In the database file, we include individual digital identification of all bibliographical sources used to populate the dataset. The database is also presented in our previous article[1]: https://doi.org/10.1039/D3NJ05784D.

The digital representation (mol file) of each chemical structure involved is also presented in the repository, in the folder “Mol Files”. The chemical species are sorted in each descriptor name: ‘_Cat’ for Catalyst, ‘_Sol’ for Solvent, ‘_Lig’ for Ligand and ‘_Sub’ for Substrate.

We also provide code for descriptor generation in the online repository.

Annex S2 – Data Augmentation Protocol

The data augmentation protocol expands the dataset while ensuring that the generated synthetic data remains within experimentally observed boundaries. I{Wei, 2022 #993}t is designed to conform to chemical plausibility and enhance the performance of the ML model. The full script is on the article's online repository.

1. Dataset Preprocessing and Constraint Definition

The dataset is first cleaned by filling missing values and converting all numeric variables to the correct format. The numeric variables are:

- Temperature (Temp_K)
- Yield (%)
- Enantiomeric Excess (EE, %)
- Substrate Quantity (Substrate_quant_mmol)
- Catalyst Quantity (Catalyst_quant_mmol)
- Ligand Quantity (Ligand_quant_mmol)
- Oxidant Quantity (Oxidant_quant_mmol)
- Additive Quantity (Additive_quant_mL)
- Solution Volume (Solution_vol_mL)
- Reaction Time (Time_h)

The dataset is then grouped by Laboratory, and for each numeric variable, the minimum and maximum values are determined within each lab's dataset. These values will be the experimental bounds that guarantee synthetic data will not exceed beyond the limits of real experimentation observations.

2. Synthetic Data Generation with Controlled Variation

For each laboratory, 20 synthetic rows are generated by randomly selecting an existing real data point and applying a variation of up to 10% to its numeric values. The adoption of a ±10% threshold for experimental variation is informed by the Gage Repeatability and Reproducibility methodology (https://asq.org/quality-resources/gage-repeatability) and the NIST Technical Note 1297 (https://www.nist.gov/pml/nist-technical-note-1297). This variation is constrained so that no synthetic value exceeds the observed minimum or maximum range for that laboratory. For

Yield (%) and Enantiomeric Excess (EE, %), additional restrictions ensure that values remain between 0% and 100% to maintain chemical relevance. Each synthetic data point is labeled with a new column (`is_synthetic = 1`), distinguishing it from the original dataset.

3. Addressing Underrepresented Yield Ranges

To ensure a balanced dataset, the Yield (%) variable is divided into 10 discrete bins based on its distribution. The number of data points in each bin is compared to the average count per bin, and bins with fewer than 50% of the average count are identified as underrepresented. 20 additional synthetic data points are generated for each underrepresented bin to create a more even distribution.

4. One-Hot Encoding of Categorical Variables

To prepare the dataset for machine learning, categorical variables (Catalyst Structure, Laboratory, Catalyst, Substrate, Ligand, Oxidant, and Solvent) are transformed using one-hot encoding. Any missing categorical values are replaced with "missing" before encoding.

5. Final Data Storage

The final dataset, now containing both real and synthetic data, is saved as for further modeling.

In Figure A1, we present the augmented reaction yield distribution for all catalysts.

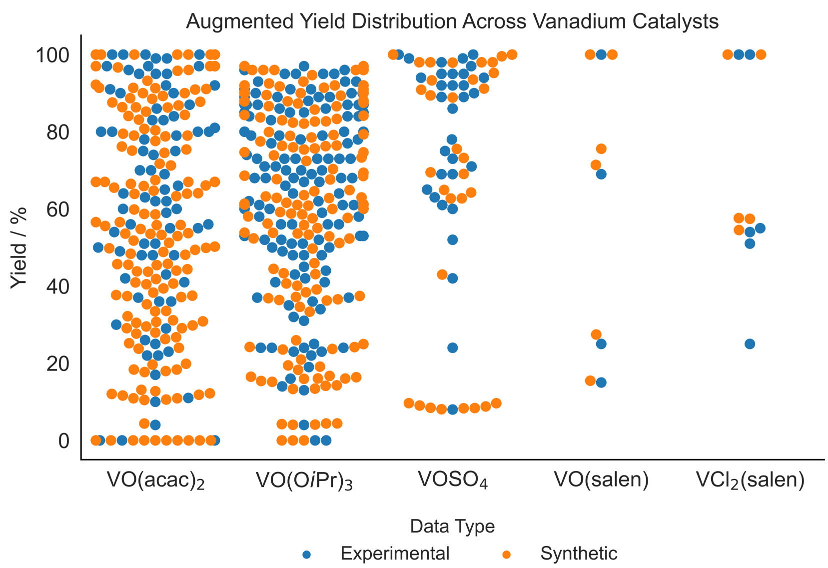

Figure A1. Reaction yield distribution of each reaction entry in the augmented model's database, based on all five represented catalyst types.

Annex S3 – Data Augmentation Distribution Analysis

To support the argument of Data Augmentation (DA) for our ESA dataset, t-SNE (t-distributed Stochastic Neighbor Embedding) plots can be instrumental in visualizing how the augmented data represents the original dataset. The t-SNE plot is a powerful, by its non nonlinear method for the visualization of high-dimensional data by projecting back into two-dimensional or three-dimensional plots. They are useful for revealing hidden relations, patterns, and clusters in complex datasets that are usually very hard to understand. By comparing the t-SNE plots of the datasets before and after DA, we will assess whether the augmentation succeeded in extending the coverage of the datasets and filling in underrepresented areas. Figure A2 represents the t-SNE plots before and after DA. A two-component separation was used as it proved to be the best score on data distribution for the DA dataset (2 components = 0.6917; 3 components = 0.6449).

This chart provides evidence that supports data augmentation in our model. In the original dataset, there is a clear clustering of certain catalysts on the t-SNE plots, like $VO(acac)_2$ and $VO(O\textit{i}Pr)_3$, suggesting that the dataset may be biased towards particular catalysts, with some regions of chemical space underrepresented. After applying data augmentation, the second plot demonstrates a more balanced and expanded coverage of the chemical space, with the catalyst structures more evenly distributed. Although the KNN preservation score drops in the augmented dataset, this suggests that the model sacrifices precision in nearest-neighbor relationships for a broader representation of the chemical space. Such trade-offs render the model better capable of generalizing across other catalysts.

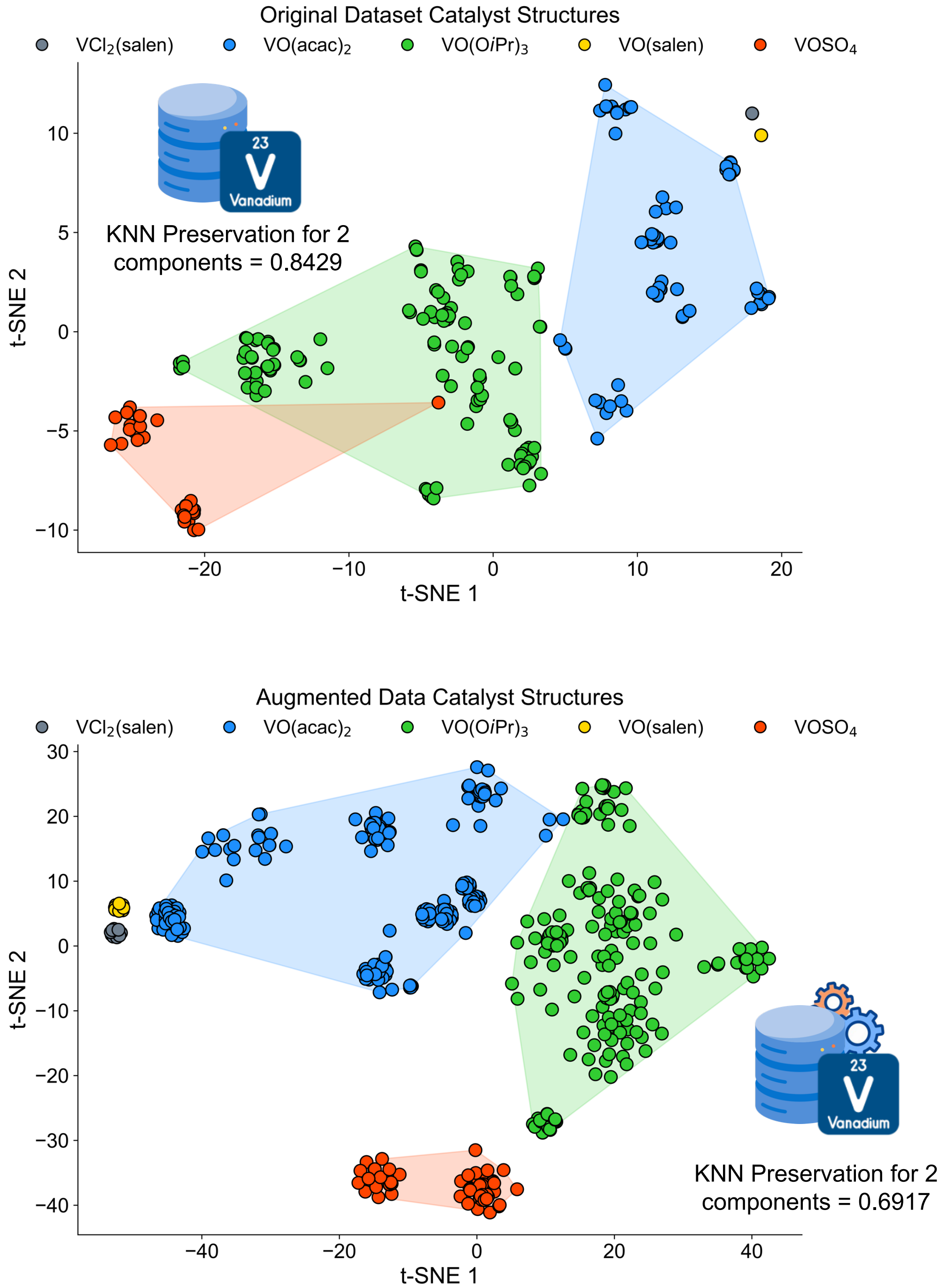

Figure A2. ESA data distribution t-SNE plots before and after DA using 2 components.

Annex S4 – Results and Model Optimization Features

Best hyper-parameters for each algorithm for model optimization:

Random Forest Regressor = n_estimators= 150, min_samples_split= 3, min_samples_leaf= 1, max_features='sqrt', max_depth=None, bootstrap= False, random_state=47

Gradient Boosting Regressor = n_estimators=150, min_samples_split=4, min_samples_leaf= 3, max_depth=40, random_state=47

Cross validation scoring result

CV = RepeatedKFold(n_splits=2, n_repeats=3, random_state=47)

Scoring Mode = 'neg_mean_absolute_error', cv=cv, n_jobs=-1, error_score='raise'

Cross Validation STD Score RF Optimized Model = 1.7 ± 0.8 (Average MAE = 4.7 ± 0.3)

Cross Validation STD Score GB Optimized Model = 1.9 ± 0.9 (Average MAE = 4.3 ± 0.7)

The scripts for algorithm development and hyperparameter optimization are in the article's online repository, in the "Scripts" folder.

Results Description

The full statistical results are available in the "Results" folder of the online repository. The file "ESA-Model-Aug-Full-Results.xlsx" contains the full results, organized in separate sheets:

- *Model Training Size Results:* Initial pre-optimization R^2 ESA yield prediction performance with increasing training load for all tested algorithms, as depicted in Figure 5 of the Manuscript.
- *Opt RF Results - Cat Type:* ML model epoxidation yield prediction performance with optimized RF algorithm. The 'Cat_Structure' column indicates what types of target

catalysts are predicted. ‘Cat_Structure = All’ column presents the individual results for the Table 1 of the manuscript.

- *Opt GB Results - Cat Type:* ML model epoxidation yield prediction performance with optimized GB algorithm. The ‘Cat_Structure’ column indicates what types of target catalysts are predicted. ‘Cat_Structure = All’ column presents the individual results for the Table 1 of the manuscript.
- *Feature Importance RF Opt Model:* Feature importance results for the descriptors in ML model for ESA yield prediction using the RF algorithm, sorted by descriptor type and source. This data is presented in Figure 11 of the manuscript.
- *Feature Importance GB Opt Model:* Feature importance results for the descriptors in ML model for ESA yield prediction using the GB algorithm, sorted by descriptor type and source.
- *Top 15 FI Score:* ML model epoxidation yield prediction performance with optimized RF algorithm, using only the top 15 descriptors form Feature Importance of the optimized RF algorithm using all descriptors. Results are considered for all catalysts in bulk.
- *RF Opt - DA Weight Test:* Error metrics for ESA yield prediction with varying synthetic data weights and variation of test R^2 scores for ESA yield prediction with RF algorithm with different synthetic data weights. Figure 10 of the manuscript presents two charts for results visualization.

Full Descriptor List (n=688) is depicted in the Feature Importance result sheets (*Feature Importance RF/GB Opt Model).*

Final role of descriptors after model processing (n=688)

For the final descriptor list amassed after processing: (i) 106 x 4 RDKit descriptors per catalyst, ligand, substrate and oxidant n=424); (ii) one-hot encoding of names of substrate, solvent, catalyst, ligand, oxidant, catalyst structure, laboratory (author) n=250; (iii) solvent Minnesota descriptors n=8; descriptors with experimental information (Additive_quant_mL, Batch Variation, is_synthetic, Solution_vol_mL, Temp_K, Time_h) n=6.

In Figure A3, we present the SHAP summary plot showing the contribution of the top 20 features to the model's prediction of reaction yield. Each point represents a single data instance, with position along the x-axis indicating the SHAP value (feature impact on the model).

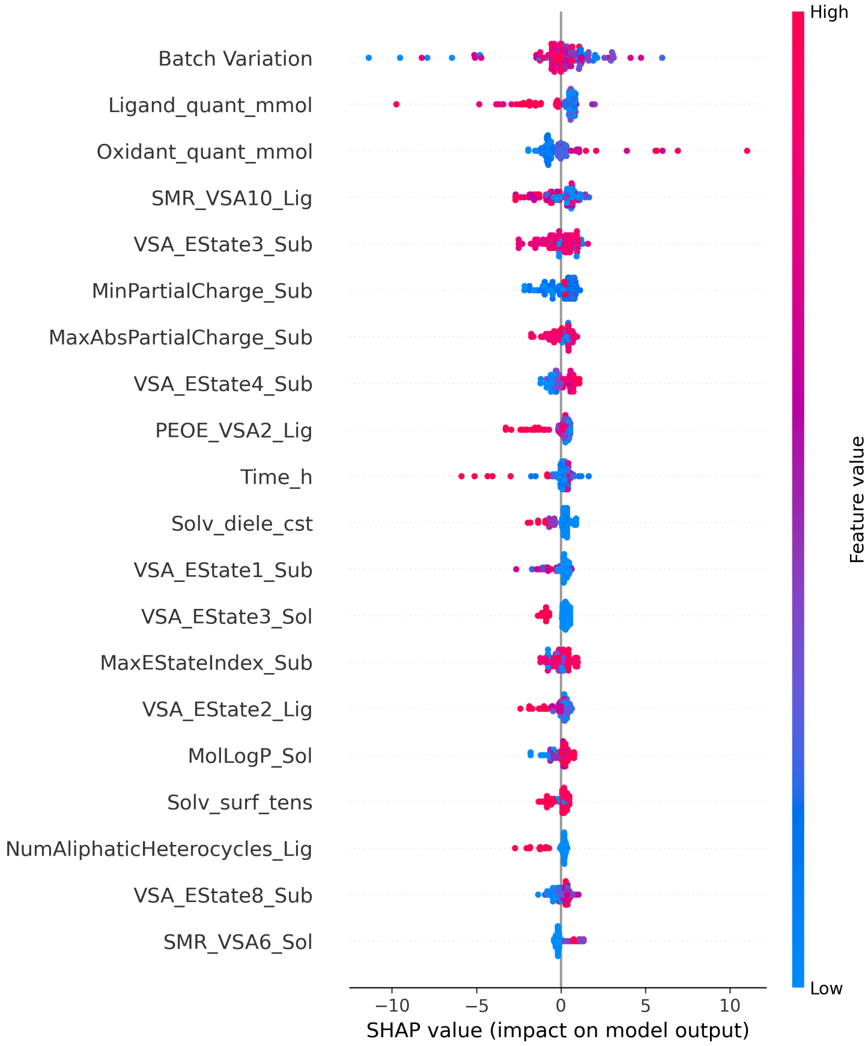

Figure S3. SHAP summary plot showing the contribution of the top 20 features to the model's prediction of reaction yield. Color reflects the magnitude of the feature value (blue = low, red = high). Features are ordered by their overall importance, with those at the top exerting the greatest average influence on model predictions.

We have also analyzed descriptor relationships using a correlation heatmap, presented in Figure S4, which reveals that most prominent descriptors are not significantly correlated with one another.

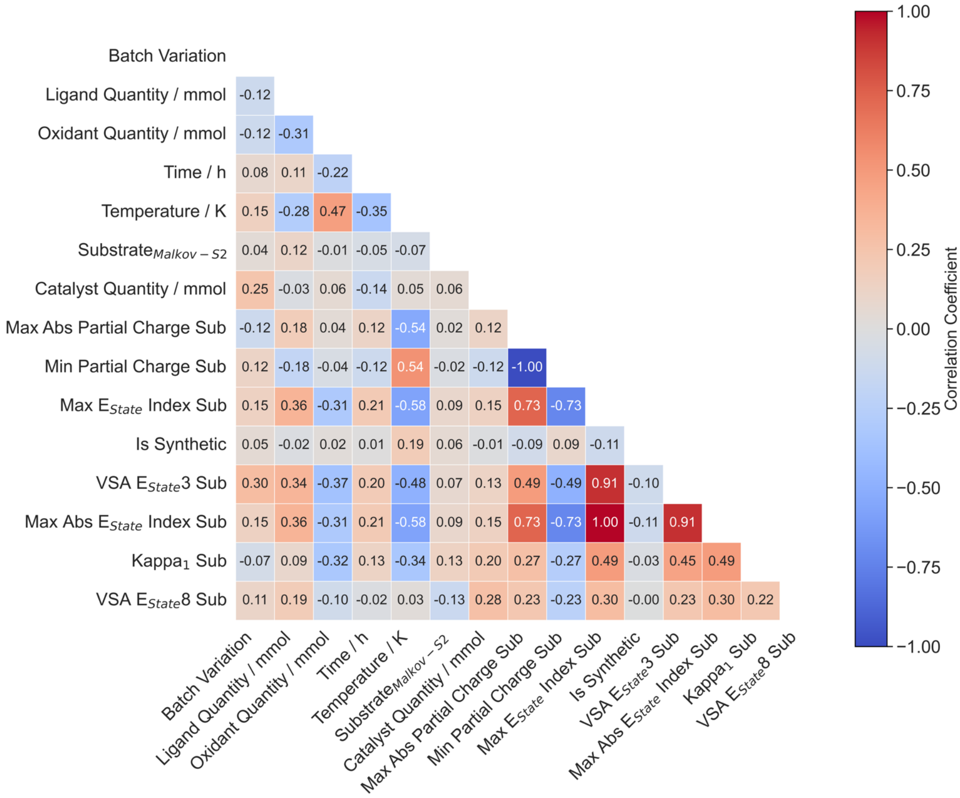

Figure S4. Correlation heatmap for the top 15 descriptors with higher FI in the ML optimized model's ESA yield predictions.

Annex S5 – RDKit Full Descriptor List

Descriptors used in Machine Learning (ML) model development (Table S1) retrieved from RDKit software. The respective bibliographical foundation of each descripted is described in detail in the RDKit WebBook Documentation[2]. The descriptor calculations were made by converting each SMILES string (representing each molecular entry), running on top of Python version 3.9[3] and using RDKit package, version 2022.09.4[4]. The Chemical information Type is presented in Table S1, and the chemical species are sorted in the database file in the online repository. Note that some entries represent multiple related descriptors.

Table S1

List of RDKit Descriptors used in ML model development (n = 106 descriptors).

Descriptor	Chemical information Type	Meaning
BalabanJ (1)	Structural	Distance sum of the two end-vertex for each edge. Connectivity of a molecule based on its graph structure (BalabanJ index has been proven to be relevant to network branching).
BertzCT (1)	Structural	Complexity index, considering both the variety of kinds of bond connectivity's and atom types; information contents related to bond connectivity and atom type diversity.
0χ , 1χ (2)	Structural	This descriptor signifies a retention index derived directly from gradient retention times. Considers individual atoms and their valence (0) and pairs of directly connected atoms (1).
0χn – 4χn (5)	Structural	This descriptor signifies a retention index derived directly from gradient retention times (normalized indexes).
0χv – 4χv (5)	Structural	This descriptor signifies atomic valence connectivity index. Indices weighted by atomic valence
EState_VSA1 – EState_VSA11 (11)	Surface Area	MOE-type (QSAR model) descriptors using EState indices and surface area contributions. RDKit automatically calculates the EState index for each atom in the molecule, assigning each atom to a

		predefined EState range (bin). Sums the VSA of atoms within each bin (surface area grouped by EState range).
Hall Kier α (1)	Structural	Descriptor fingerprint that displays the difference between active and inactive molecules.
HeavyAtomCount (1)	Structural	The number of heavy atoms in the molecule.
HeavyAtomMolWt (1)	Structural	The average molecular weight of the molecule ignoring hydrogens.
κ_1 , κ_2 , κ_3 **(3)**	Structural	This descriptor signifies # κ shape index: $(n\text{-}1) \times 2 / m^2$
***Max and Min AbsEStateIndex* (2)**	Electronic	Maximum and minimum absolute E-State
***Max and Min AbsPartialCharge* (2)**	Electronic	Maximum and minimum absolute partial charge
***Max and Min EStateIndex* (2)**	Electronic	Maximum and minimum E-State
***Max and Min PartialCharge* (2)**	Electronic	Maximum and minimum partial charge
Mol logP (1)	Structural	Wildman-Crippen logP value.
MolMR (1)	Electronic	Wildman-Crippen molar refractivity.
MolWt (1)	Structural	The average molecular weight of the molecule.
NHOH Count (1)	Structural	The number of NHs or OHs.
NO Count (1)	Structural	The number of Nitrogens and Oxygens.
$n_{alicarb}$ **(1)**	Structural	The number of aliphatic carbocycles.
n_{alihet} **(1)**	Structural	The number of aliphatic heterocycles.
n_{alirig} **(1)**	Structural	The number of aliphatic rings.
$n_{arocarb}$ **(1)**	Structural	The number of aromatic carbocycles.
n_{arohet} **(1)**	Structural	The number of aromatic heterocycles.
n_{arorig} **(1)**	Structural	The number of aromatic rings.

n_{Ha} (1)	Structural	The number of Hydrogen Bond Acceptors.
n_{Hd} (1)	Structural	The number of Hydrogen Bond Donors.
n_{het} (1)	Structural	The number of Heteroatoms.
n_{radele} (1)	Structural	The number of radical electrons.
n_{rot} (1)	Structural	The number of rotatable bonds.
$n_{satcarb}$ (1)	Structural	The number of saturated carbocycles.
n_{sathet} (1)	Structural	The number of saturated heterocycles.
n_{satrig} (1)	Structural	The number of saturated rings.
n_{ele} (1)	Structural	The number of valence electrons.
PEOE_VSA1 – PEOE_VSA14 (14)	Surface Area	MOE-type (QSAR model) descriptors using partial charges estimated using the Gasteiger PEOE and surface area contributions (sum the VSA of atoms within a given partial charge range).
SMR_VSA1 – SMR_VSA10 (10)	Surface Area	MOE-type (QSAR model) descriptors using atomic contributions to molar refractivity and surface area contributions (sums the VSA of atoms in each bin).
SlogP_VSA1 – SlogP_VSA12 (12)	Surface Area	MOE-type (QSAR model) descriptors using logP (calculated using the Crippen method) contributions and surface area sum of surface areas of atoms whose logP contribution falls within a specific range).
TPSA (1)	Surface Area	The total polar surface area of a molecule based upon fragment calculations.
VSA_EState1 – VSA_EState10 (10)	Surface Area	MOE-type (QSAR model) descriptors using EState indices and surface area contributions. The sum of VSA contributions of atoms whose EState values fall into specific predefined ranges (surface area grouped by predefined VSA ranges).

Annex S6 – Experimental Descriptor List

Experimental descriptors used in ML model development (Table S2) retrieved from each bibliographical source. Specific solvent parameters were retrieved from the Minnesota Solvent Descriptor Database[5]. Each entry represents one descriptor.

Table S2. List of experimental descriptors used in machine learning model development (n = 18).

Descriptor	Chemical information Type	Meaning
Temp_K	Experimental	Reaction temperature in Kelvin.
Yield	Experimental	Reaction yield.
EE	Experimental	Reaction enantiomeric excess.
Configuration	Experimental	Product geometrical configuration.
Batch Variation	Experimental	Attributes the same value to reactions that have the same experimental procedure conducted under differing conditions.
is_synthetic	Experimental	Attributes values of 0 or 1 if reaction is experimental or from augmented strategy.
Substrate_quant_mmol	Experimental	Quantity of reaction substrate in mmol.
Catalyst_quant_mmol	Experimental	Quantity of reaction catalyst in mmol.
Ligand_quant_mmol	Experimental	Quantity of reaction catalyst ligand in mmol.
Oxidant_quant_mmol	Experimental	Quantity of reaction oxidant in mmol.
Additive_quant_mL	Experimental	Volume of reaction additive in mL.
Solution_vol_mL	Experimental	Volume of reaction solution in mL.
Time_h	Experimental	Reaction duration in hours.
Solvent	Experimental	Reaction solvent.
Solv_opt_freq20	Experimental	Solvent index of refraction at optical frequencies at 293 K.
Solv_opt_freq25	Experimental	Solvent index of refraction at optical frequencies at 298 K.

Solv_Hbond_ac	Experimental	Solvent Abraham’s hydrogen bond acidity.
Solv_Hbond_bs	Experimental	Solvent Abraham’s hydrogen bond basicity.
Solv_surf_tens	Experimental	Solvent surface tension.
Solv_diele_cst	Experimental	Solvent dielectric constant at 298 K.

Annex S7 – Sample Code for ML Model Design

The scripts for algorithm development are in the articles online repository, in the folder Scripts.

Scripts Overview

1. mol-conversion.py

This script provides scripts molecular data processing and conversion. It transforms molecular structures contained within the dataset into formats applicable for ML models. Key functionalities include:

- Converting molecular data (e.g., SMILES strings) into descriptor formats suitable for model input.
- Handling molecule-based transformations, ensuring that molecular representations are standardized and compatible with the model.
- The output of this script is typically used as input features for subsequent model training and evaluation.

2. model_preliminary_run.py

This script provides initial evaluations of ML models to deliver baseline performance metrics. The model evaluation is performed without hyperparameter tuning on a set of algorithms as a starting point to gauge how well the models perform on the initial dataset. Functions include:

- Loading the dataset and preparing it for modeling.
- Running baseline models: Random Forest, Gradient Boosting, Support Vector Machines, Neural Networks.
- Outputting preliminary performance metrics (R^2, MAE, RMSE) to help guide future model selection and optimization steps.

3. ESA_data_augmentation.py

This script is used for **data augmentation** on the ESA-Vanadium dataset. It enriches the dataset by generating synthetic data points or applying transformations to existing data. Key functions of this script include:

- Handling missing values or filling gaps in the dataset.
- Augmenting the dataset to expand feature space, enabling robust model training.
- Ensures that augmented data remains representative of the original chemical reaction conditions, focusing on maintaining chemical consistency.

4. ESA-hyperparameter-opt.py

This script performs hyperparameter optimization for a Random Forest model using the RandomizedSearchCV function from scikit-learn. The goal is to optimize the model's performance by searching over a predefined range of hyperparameters. The script:

- Loads the ESA-Vanadium dataset, excluding certain irrelevant features.
- Conducts Random Forest regression, applying a hyperparameter search over parameters like the number of estimators, depth, and minimum samples split.
- Evaluates model performance using metrics such as R^2, mean absolute error (MAE), and root mean squared error (RMSE) over multiple iterations.
- Saves the optimized model results and the best hyperparameters to a CSV file.

5. ESA_model_opt_run.py

This script focuses on the **execution and evaluation** of the optimized RF ML model on the augmented ESA-Vanadium dataset. It is responsible for:

- Collecting results from each model's training and testing phases.
- Recording performance metrics, including training and test R^2, MAE, and RMSE for each model and iteration.
- Saving the evaluation metrics and model-specific results for further analysis or comparison.

6. partial_dependency_plots.py

This script generates partial dependence plots (PDPs) to interpret model behavior. PDPs show how each feature impacts the predicted outcome, allowing for better understanding of the model's decision-making process. The script:

- Loads the trained model and dataset.
- Generates plots that display the relationship between individual features and the model's predictions.
- Visualizes feature importance and dependencies, helping to uncover key drivers in the reaction outcomes.

7. weight_analysis.py

The script records optimized RF model performance across 10 iterations with varying synthetic data weights. These weights, ranging from 0.1 to 0.9, adjust the contribution of synthetic reactions in the training data, simulating different levels of experimental noise.

8. 'top_15_fi_RF.py'

Script to run the optimized RF model using the top 15 features based on Feature Importance of model prediction using all catalysts.

- Returns a csv with the model’s predictions.

9. 'shap_analysis.py'

Same framework of the execution and evaluation of the optimized RF machine learning model on the augmented ESA-Vanadium dataset, while also yielding the SHAP analysis for the “All Catalysts” case.

- Returns a csv with the model's SHAP values.
- Generates a SHAP plots that display feature influence in the final model’s predictions.

References

(1) Ferraz-Caetano, J.; Teixeira, F.; Cordeiro, M. N. D. S. Navigating epoxidation complexity: building a data science toolbox to design vanadium catalysts. *New Journal of Chemistry* **2024**, *48* (12), 5097-5100, 10.1039/D3NJ05784D. DOI: 10.1039/D3NJ05784D.

(2) *RDKit: Open-source cheminformatics - Descritptor Webbook*. https://www.rdkit.org/docs/GettingStartedInPython.html#list-of-available-descriptors (accessed January 18th, 2024).

(3) *Python Software Foundation - Python Language Reference, version 3.9.8*. http://www.python.org (accessed January 18th, 2024).

(4) Landrum, G. *RDKit: Open-source cheminformatics 2022_09_4 (Q3 2022) Release - January 16, 2023*. 2023. http://www.rdkit.org/ (accessed January 18th, 2024).

(5) Marenich, A. V. K., Casey P; Thompson, Jason D; Hawkins, Gregory D; Chambers, Candee C; Giesen, David J; Winget, Paul; Cramer, Christopher J; Truhlar, Donald G. *Minnesota Solvation Database (MNSOL) version 2012. Retrieved from the Data Repository for the University of Minnesota*. 2020. https://doi.org/10.13020/3eks-j059 (accessed January 18th, 2024)